

CFD ile HİSTOGRAM TABLOSU OLUŞTURMA ADIMLARI:

- 1- Veri aralığını hesapla (son veri - ilk veri)
- 2- 5-15 arasından bir k sayısı belirle (3.formülden birini de kullanabilirsin.)
- 3- Aralığı k'ye bölerek sınıf aralığını bul
- 4- Tüm sınıfların orta noktasını bul
- 5- Sınıflarda yer alan data sıklığını (frequency) bul

SIKLIK POLİGONU TABLOSU OLUŞTURMA:

Histogram tablosu oluştururken belirlediğimiz orta noktalardan düz çizgilerle birleştiririz.

$$k = \sqrt{n}$$
$$k = 1 + 3.3 \log n$$
$$k = 5 \log n$$

NOT: Histogramda dörtgen altında kalan alan yani sıklık*sınıfaralığı bütün aralıklarda orantılı olmalıdır. Eğer sınıf aralığını büyütürsek ona göre sıklığın değeriyle oynamalıyız.

STATISTICS

PART 1: Basic Concepts, Graphical Projection of Data, Frequency Distributions

Birol Aslanyürek

Rev:27.09.2024

Basic Concepts of Statistics

Statistics

2

Engineering, scientific, industrial, etc. problems can be solved through

Analytical Methods

(Exact or approximate solutions)

Numerical Methods

(Approximate solutions)

Statistical Methods or Artificial Intelligence Methods

(Estimated solutions)

Deterministic Approach

The problem is investigated under specific conditions.

Stochastic Approach

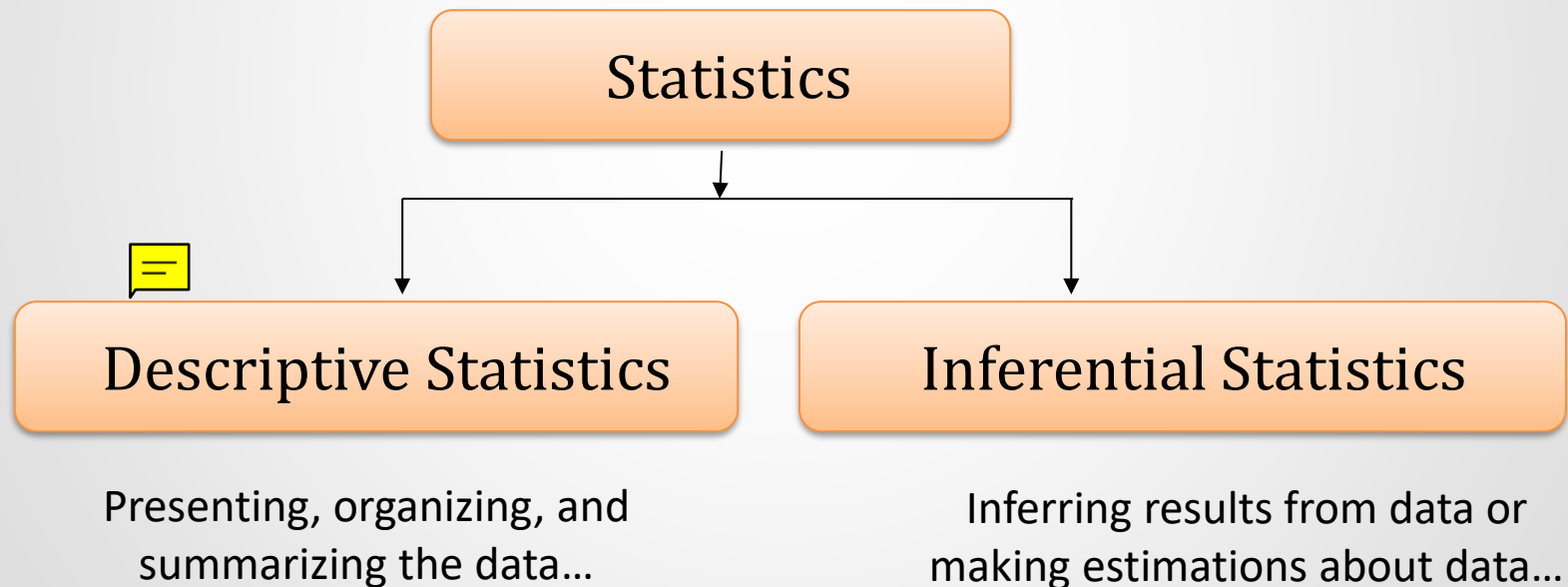
The problem is investigated under probabilistic conditions by using collected data.

Basic Concepts of Statistics

Statistics

3

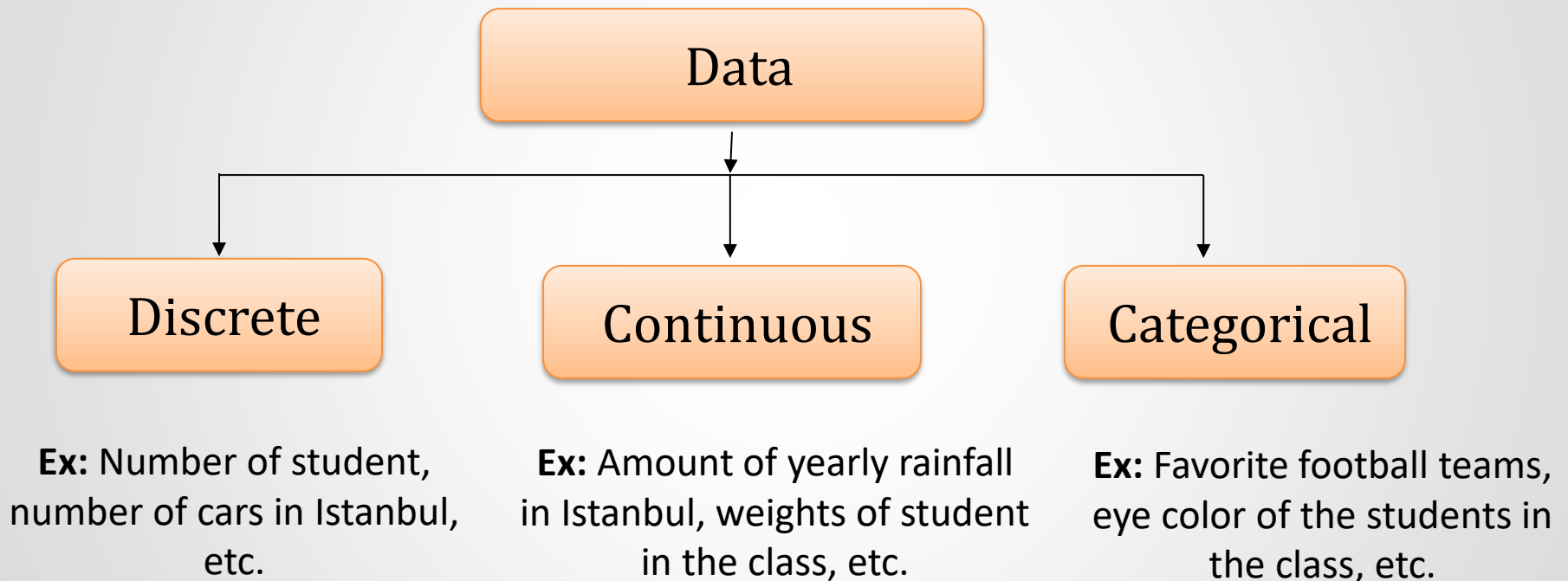
Definition: **Statistics** is the science of collection, organization, interpretation, and estimation of data.



Basic Concepts of Statistics

Statistics

4



Graphical Projection of Data

Statistics

5

Consider the following illustrative example...

EXAMPLE 1 (PUZZLE EXAMPLE): In order to learn in how many minutes a newly designed puzzle can be solved, 50 children were observed when they solved this puzzle. The following table shows solving times (in minutes) for the puzzle:

4	15	6	11	2	15	9	21	13	10
17	19	11	8	11	13	12	8	17	9
15	24	14	7	13	16	14	12	15	14
5	10	13	23	12	31	13	25	5	16
11	15	26	17	16	14	28	10	15	19

Graphical Projection of Data

Statistics

6

How can we describe the given data?

4	15	6	11	2	15	9	21	13	10
17	19	11	8	11	13	12	8	17	9
15	24	14	7	13	16	14	12	15	14
5	10	13	23	12	31	13	25	5	16
11	15	26	17	16	14	28	10	15	19



The simplest organization of the data is sorting them.

2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

Graphical Projection of Data

Statistics

7

The Sorted Data

2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

Min. value

$n = 50$

Max. value

$$\begin{aligned}\text{Range of the data} &= \text{Max. Value} - \text{Min value} \\ &= 31 - 2 \\ &= 29\end{aligned}$$

Graphical Projection of Data

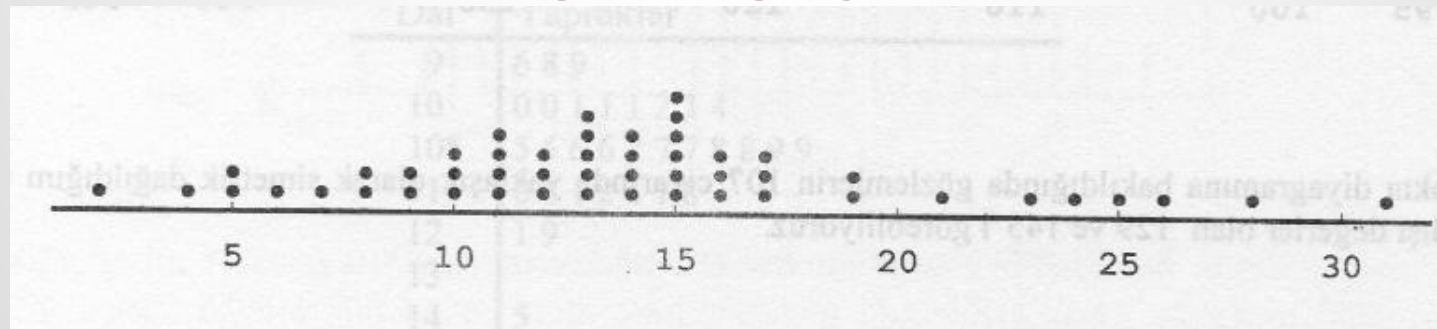
Statistics

8

2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

There are many different ways to illustrate data.

DOT PLOTS



Dot plots can be used to see the distribution of data, where data are concentrated or extreme values, etc.

Graphical Projection of Data

Statistics

9

2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

STEM-AND-LEAF PLOTS

Stem	Leaf
0	2 4 5 5 6 7 8 8 9 9
1	0 0 0 1 1 1 1 2 2 2 3 3 3 3 3 4 4 4 4 5 5 5 5 5 5 6 6 6 7 7 7 9 9
2	1 3 4 5 6 8
3	1

Since the given data set consists of one and two-digit numbers, here stem represents the tens-digit of the values while the leaf represents the ones-digit of the values.

Graphical Projection of Data

Statistics

10

2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

In order to obtain more illustrative stem-and-leaf plots, we can increase the number of stems.

Stem	Leaf
0	2 4
0*	5 5 6 7 8 8 9 9
1	0 0 0 1 1 1 1 2 2 2 3 3 3 3 3 4 4 4 4
1*	5 5 5 5 5 5 6 6 6 7 7 7 9 9
2	1 3 4
2*	5 6 8
3	1

How can stem-and-leaf plots be constructed for the data with non-integer values?

Graphical Projection of Data

Statistics

11

2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

FREQUENCY DISTRIBUTION

Time (min)	Frequency (f)	Time (min)	Frequency (f)
2	1	14	4
4	1	15	6
5	2	16	3
6	1	17	3
7	1	19	2
8	2	21	1
9	2	23	1
10	3	24	1
11	4	25	1
12	3	26	1
13	5	28	1
		31	1

Another way of showing the frequency of values in a data set is using a **frequency distribution (FD) table**.

$$\sum_i f_i = n = 50$$

Graphical Projection of Data

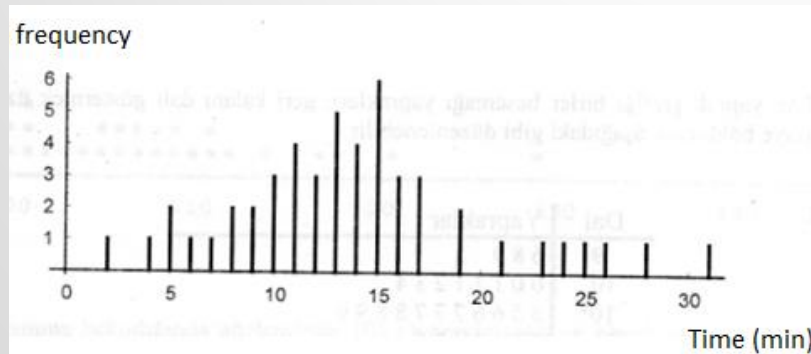
Statistics

12

LINE PLOT

Time (min)	Frequency (f)	Time (min)	Frequency (f)
2	1	14	4
4	1	15	6
5	2	16	3
6	1	17	3
7	1	19	2
8	2	21	1
9	2	23	1
10	3	24	1
11	4	25	1
12	3	26	1
13	5	28	1
		31	1

For the graphical display of frequency distribution tables, we often use **line plots**.



Graphical Projection of Data

Statistics

13

CLASSIFIED FREQUENCY DISTRIBUTION AND HISTOGRAMS

When a data set contains continuous data or a large number of discrete data, it may not be appropriate to use **FD**. For such cases, we can use **classified frequency distribution (CFD)**.

Steps for Creating CFD:

1. Compute the range of data.
2. Determine the number of classes (k). In practice, k can be chosen generally between 5-15. You may use the following formulas to specify it:

$$k = \sqrt{n}$$
$$k = 1 + 3.3 \log n$$
$$k = 5 \log n$$

Graphical Projection of Data

Statistics

14

CLASSIFIED FREQUENCY DISTRIBUTION AND HISTOGRAMS

3. Divide the data into classes with equal intervals. To create these classes, first compute the class interval:

$$\text{Class interval} = \left[\frac{\text{Range of data}}{\text{Number of classes } (k)} \right]$$

4. Find the midpoints (x_i) of each class.
5. Compute the frequency (f_i) of each class.

Graphical Projection of Data

Statistics

15

Ex: Create a classified frequency distribution table for the puzzle example.

2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

(1) Range = 29

(2) For the given data set, we choose k as 6. (Here we have not used the given formulas.)

$$(3) \text{ Class interval} = \left\lceil \frac{29}{6} \right\rceil = 5$$

Graphical Projection of Data

Statistics

16

2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)
2-7	4,5	5
7-12	9,5	12
12-17	14,5	21
17-22	19,5	6
22-27	24,5	4
27-32	29,5	2
(4)	(5)	$\sum_{i=1}^k f_i = 50$

Attention: In this course, we will assume that the value at the lower bound of a class belongs to that class, while the value at the upper bound does not.

For example, the value '7' belongs to the second class (7-12).

Graphical Projection of Data

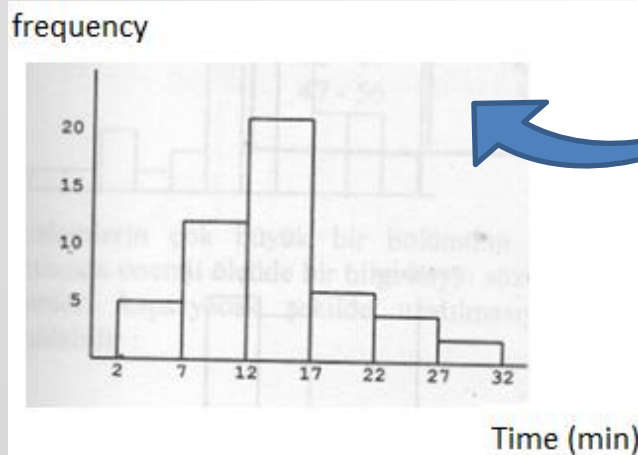
Statistics

17

HISTOGRAMS

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)
2-7	4,5	5
7-12	9,5	12
12-17	14,5	21
17-22	19,5	6
22-27	24,5	4
27-32	29,5	2

For the graphical display of CFD tables, we use **histograms** instead of line plots.



- ❖ Here, rectangles represent the classes or mid-points.
- ❖ Histograms are not same with bar graphs!
- ❖ The area of each rectangle is proportional to the corresponding frequency.

Graphical Projection of Data

Statistics

18

CLASSES WITH UNEQUAL INTERVALS

Assume that 4 new data points are added to the data set: 28, 34, 41 and 55.

In this case;

$$\text{Range} = 55 - 2 = 53$$

If we select k as 6 again,

$$\text{Class interval} = \left\lceil \frac{53}{6} \right\rceil = 9$$

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)
2-11	6,5	13
11-20	15,5	30
20-29	24,5	7
29-38	33,5	2
38-47	42,5	1
47-56	51,5	1
		Sum : 54

As seen from the CFD table, most of the data are concentrated in the second class. This results in a significant loss of information, specifically the details of data distribution). The issue here arises due to extreme values such as 41 and 55.

Graphical Projection of Data

Statistics

19

CLASSES WITH UNEQUAL INTERVALS

To overcome this problem, we can choose class intervals as 5 again, then extend the interval of the last class such as given below:

Classes (C_i)	Frequency (f_i)
2-11	13
11-20	30
20-29	7
29-38	2
38-47	1
47-56	1

For classes with
equal interval

Classes (C_i)	Frequency (f_i)
2-7	5
7-12	12
12-17	21
17-22	6
22-27	4
27-57	6

For classes with
unequal interval

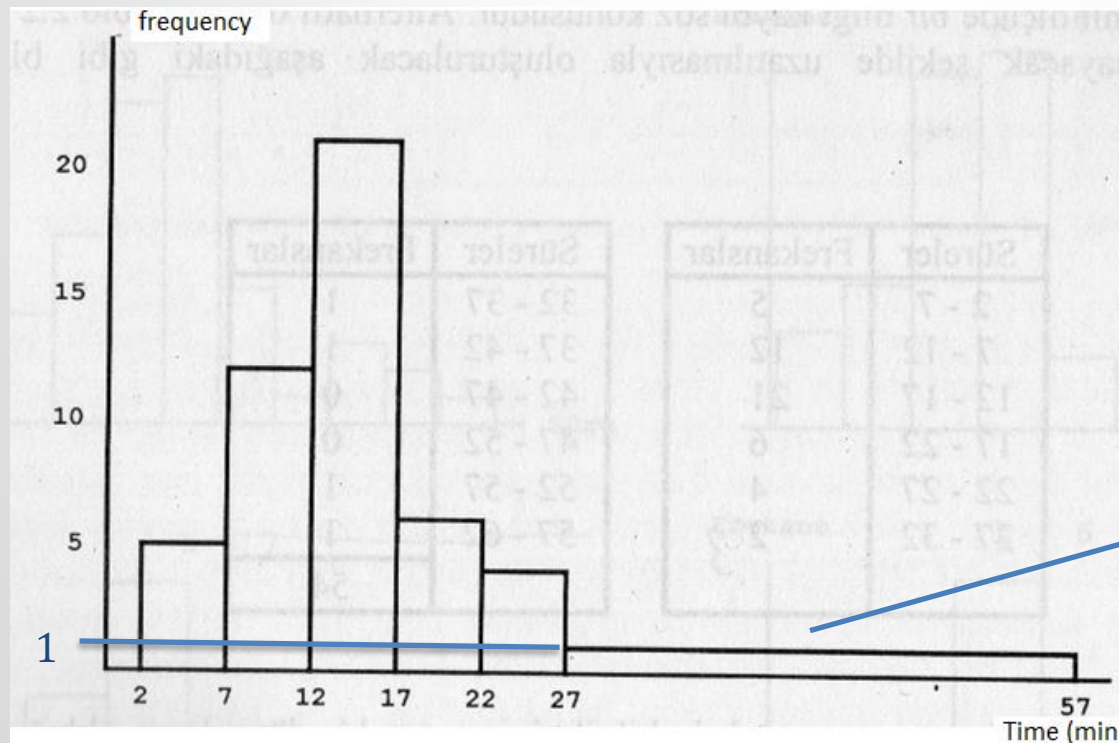
Graphical Projection of Data

Statistics

20

HISTOGRAM FOR CLASSES WITH UNEQUAL INTERVALS

The area of each rectangle in a histogram is equal to the corresponding frequency if the class intervals (width of rectangles) are normalized with 1. In this case, the sum of rectangle areas will be the number of data, i.e., n .



Why is the height of the rectangle 1 instead of 6?

Graphical Projection of Data

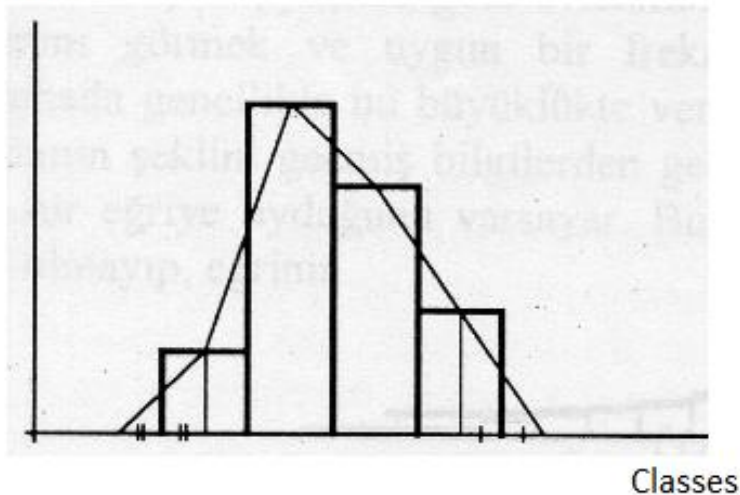
Statistics

21

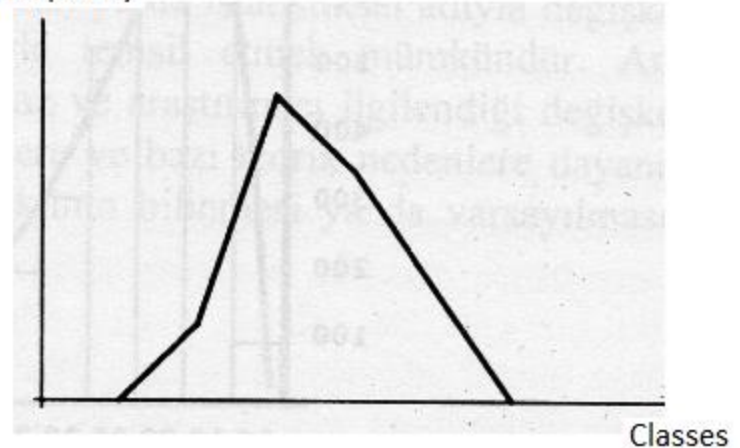
FREQUENCY POLYGON

A frequency polygon consists of the lines between midpoints of rectangles in a histogram. The first and the last line intersect the horizontal axis. The area below the frequency polygon is equal to the total area of rectangles.

Frequency



Frequency



Frequency Polygon

Graphical Projection of Data

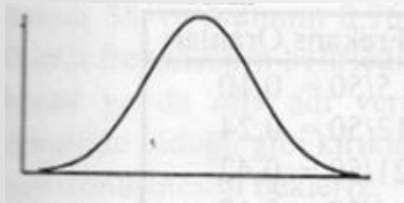
Statistics

22

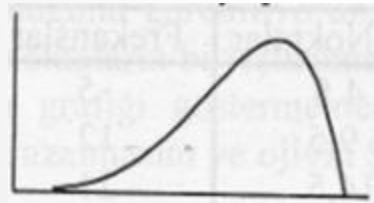
FREQUENCY CURVE

When the number of data (accordingly the number of classes) increases, the frequency polygon gets smoother. In this case, this polygon is called a **frequency curve**.

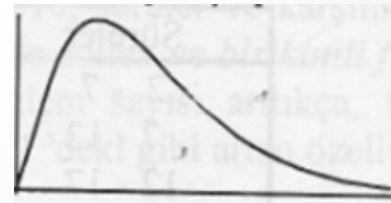
Some Basic Types of Frequency Curves



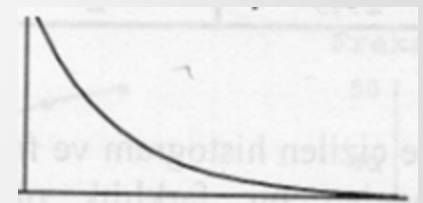
Symmetric



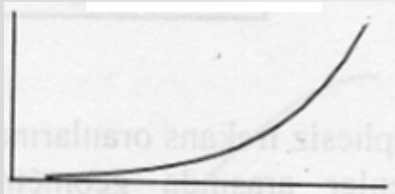
Skewed to left



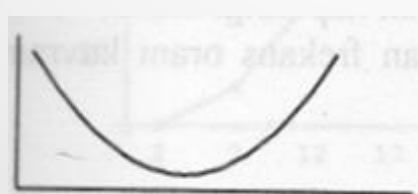
Skewed to right



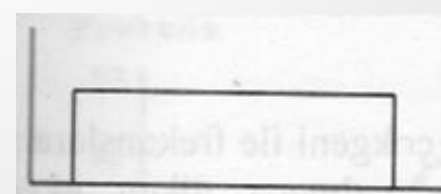
Inverse J-Shaped



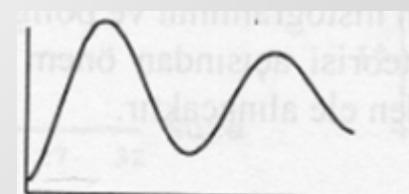
J-Shaped



U-Shaped



Rectangle



Two-mode

Graphical Projection of Data

Statistics

23

RELATIVE FREQUENCY DISTRIBUTION

The **relative frequency** of a class i , $i=1,2,...,k$, can be computed as:

$$\text{Relative Frequency } (f_i^R) = \frac{\text{Frequency } (f_i)}{\text{Number of data } (n)}$$

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)	Relative Frequency (f_i^R)
2-7	4,5	5	5/50=0,10
7-12	9,5	12	12/50=0,24
12-17	14,5	21	21/50=0,42
17-22	19,5	6	6/50=0,12
22-27	24,5	4	4/50=0,08
27-32	29,5	2	2/50=0,04

$$\sum_i f_i^R = 1$$

Note: For relative frequency, the area of the histogram is 1 if the width of each rectangle is 1. Accordingly, the area under the frequency polygon or curve is 1.

Discuss the relationship between this fact and probability functions!!!

Graphical Projection of Data

Statistics

24

(INCREASING) CUMULATIVE FREQUENCY DISTRIBUTIONS

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)	Relative Frequency (f_i^R)	(Increasing) Cumulative Frequency ($F_i \nearrow$)	Cumulative Relative Frequency ($F_i^R \nearrow$)
2-7	4,5	5	$5/50=0,10$	5	0,10
7-12	9,5	12	$12/50=0,24$	17	0,34
12-17	14,5	21	$21/50=0,42$	38	0,76
17-22	19,5	6	$6/50=0,12$	44	0,88
22-27	24,5	4	$4/50=0,08$	48	0,96
27-32	29,5	2	$2/50=0,04$	50	1,00

Graphical Projection of Data

Statistics

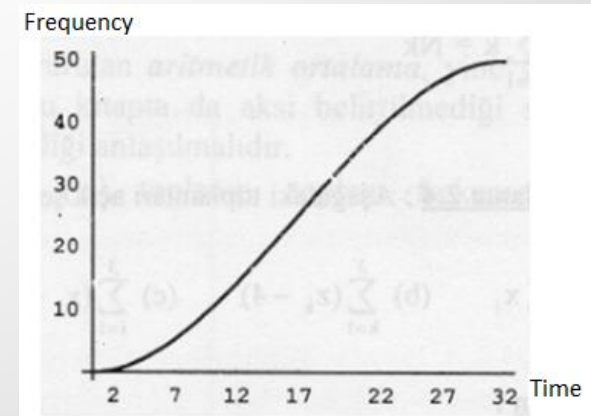
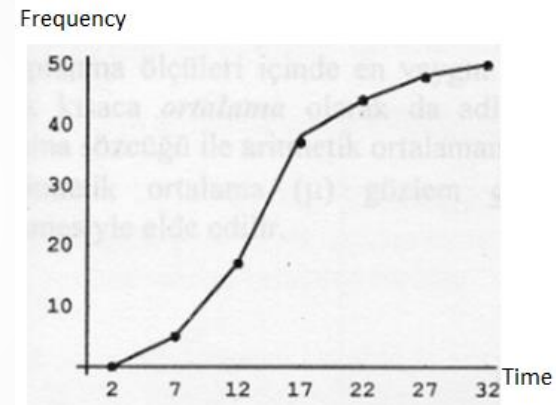
25

CUMULATIVE FREQUENCY POLYGON AND CURVE

Classes (C_i)	(Increasing) Cumulative Frequency ($F_i \nearrow$)
2-7	5
7-12	17
12-17	38
17-22	44
22-27	48
27-32	50

Cumulative
Freq. Polygon

Cumulative
Freq. Curve



Graphical Projection of Data

Statistics

26

(DECREASING) CUMULATIVE FREQUENCY DISTRIBUTION

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)	Relative Frequency (f_i^R)	(Decreasing) Cumulative Frequency ($F_i^{\searrow}$)	Cumulative Relative Frequency ($F_i^{R \searrow}$)
2-7	4,5	5	$5/50=0,10$	50	1,00
7-12	9,5	12	$12/50=0,24$	45	0,90
12-17	14,5	21	$21/50=0,42$	33	0,66
17-22	19,5	6	$6/50=0,12$	12	0,24
22-27	24,5	4	$4/50=0,08$	6	0,12
27-32	29,5	2	$2/50=0,04$	2	0,04

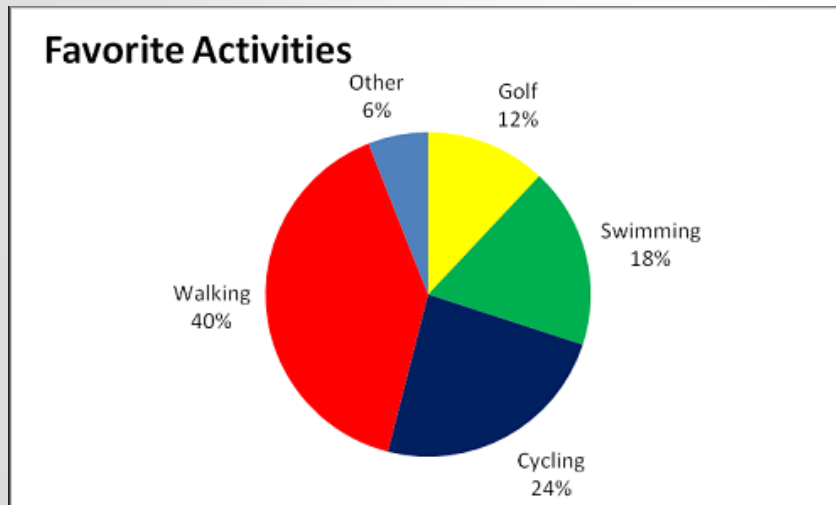
Graphical Projection of Data

Statistics

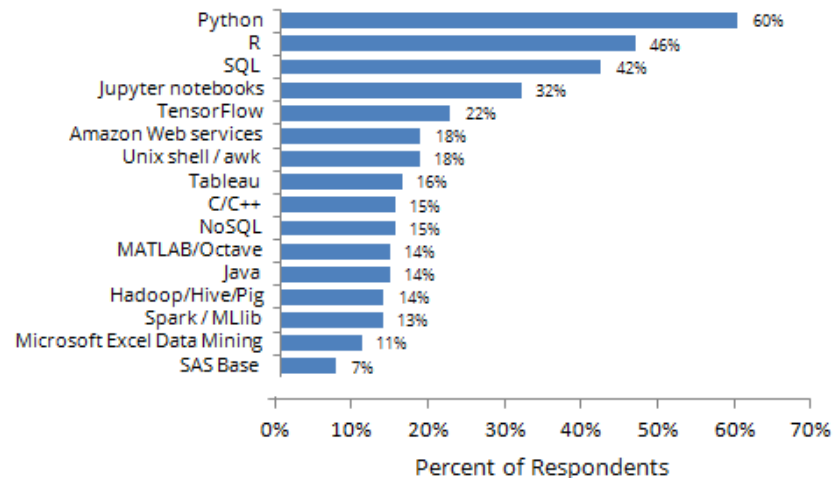
27

For the projection of numerical data, we investigated dot plots, line plots, histograms, frequency polygons, etc. When data are categorical, we can use many figures such as circular (pie) charts and bar charts.

Data science tools, technologies, languages used in 2017



A Circular Chart (1)



A Bar Chart (2)